



PMSCS PROGRAM

Department of Computer Science and Engineering
Jahangirnagar University

Date:
03/01/2026

Course Code: PMSCS-623P

Title: Data Structures and Algorithms

Time: 3 Hours

TERM: FALL 2025

Full Marks: 60

Answer any SIX from the following Questions

1.	a) Let's say you have a string "DATA". We like to reverse this string. Which data structure will you use (stack / queue) to reverse the string? Finally, perform the task using your selected data structure. b) A pharmacy receives medicine batches throughout the day. To prevent selling expired medicine, which data structure should you suggest along with the required operations for easy stock management? Explain briefly about your choice. c) Consider the following input: 5, 4, 3, 2, 1, 1, 2, 3, 4, 5. Should you apply normal Quicksort or Randomized Quicksort algorithm to sort the given input? Justify your choice.	[3+ 3+ 4]
2.	a) The MergeSort algorithm we have learned is called 2-way MergeSort because it divides the given array into two almost equal sub-arrays. In 5-way MergeSort, the given array is divided into 5 equal (almost) sub-arrays and they are recursively sorted and then combined/merged to get the sorted array. Represent the time complexity of 5-way MergeSort algorithm in Big-Oh notation. (Hint: The combine step merges a total of n elements, taking $O(n)$ time.) b) What is the number of comparisons required by Insertion Sort for a reverse-sorted array of size $(n-1)$? c) After the second pass of Bubble Sort, which element is guaranteed to be in its final position? d) A library management system needs to sort a list of books by their ID numbers whenever new books are added. Suppose the library has book IDs: [101, 102, 105, 103, 104]. Your task is to choose the most efficient sorting algorithm between Insertion Sort and Bubble Sort and justify the choice based on number of operations required to perform the task.	[4+ 2+ 1+ 3]
3.	a) Consider the array having elements as: [7, 7, 7, 7, 7, 7, 7]. Do you think linear search perform better than binary search algorithm in this case? Justify your answer. b) A giant pizza has 512 slices, and you want to share it equally among 8 friends. You can only cut the pizza in half at a time. Using a divide and conquer strategy, how many cuts do you need to make? c) Consider the following array representation of a max-heap as [90, 15, 10, 7, 12, 2, 5]. Is this a valid max-heap? Justify your answer. d) What is procedure to restore the property of a max-heap?	[3 + 2 + 3 + 2]
4.	a) Modern web browsers allow users to navigate between web pages using the Back and Forward buttons. The challenge is to always return the user to the most recently visited page first when the Back button is clicked. Which data structure should the browser use to solve this problem and why? b) To solve same problem, the performance of the two algorithms are: Algorithm A: $f(n) = 1000n$ and Algorithm B: $g(n) = n^2$. Which is better for very small n? Which is better asymptotically? Explain briefly.	[2+ 4+ 2+2 1]

	c) Which of the following is faster? i. $O(\log n)$ ii. $O(\log n^2)$ iii. $O(2 \log n)$ d) Given an array as [5, 8, -14, 27, 99, -106, 65]. Do you think that accessing the 6th element of this array will be $O(1)$?																
5.	a) How many edges will a tree have which contains total $(2n^2 + 3n - 4)$ nodes? b) Consider an undirected disconnected graph where every edge has same weight value of 2 . What information can we get if we apply BFS (Breadth-First Search) algorithm on this graph? c) BFS will always visit every node of a tree. (True / False) – Justify your answer. d) Is the Minimum Spanning Tree always unique? – Justify your answer. e) What happens if one extra edge is added to a Minimum Spanning Tree?	[2+ 2+ 2+ 2+ 2]															
6.	a) Why are more frequent characters higher in the tree in Huffman coding? b) Why must Huffman codes satisfy the prefix property? c) construct the Huffman code for the following given symbols and frequencies: <table border="1"><tr><td>Symbol</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td></tr><tr><td>Frequency</td><td>13</td><td>45</td><td>12</td><td>10</td><td>12</td></tr></table>	Symbol	A	B	C	D	E	Frequency	13	45	12	10	12	[2+ 2+ 6]			
Symbol	A	B	C	D	E												
Frequency	13	45	12	10	12												
7.	a) For activity/interval scheduling problem in greedy approach, why <i>is the earliest finish time approach</i> chosen as the best strategy? (b) Consider the following coin system having coins as {1, 5, 6, 9} and the amount that we want to exchange is 11. Do you think greedy approach will fail here? Justify your answer. c) Compute the optimal price for the 0-1 knapsack problem for knapsack capacity, W=7. Which items will be in the optimal solution? What will be their total price? Show how you calculate the optimal solution using a matrix containing both price and arrows for each cell in the matrix. <table border="1"><tr><td>Item</td><td>Weight</td><td>Value</td></tr><tr><td>1</td><td>1</td><td>1</td></tr><tr><td>2</td><td>3</td><td>4</td></tr><tr><td>3</td><td>4</td><td>5</td></tr><tr><td>4</td><td>5</td><td>7</td></tr></table>	Item	Weight	Value	1	1	1	2	3	4	3	4	5	4	5	7	[2+ 2+ 6]
Item	Weight	Value															
1	1	1															
2	3	4															
3	4	5															
4	5	7															
8.	a) How many stack push operations are required to evaluate the postfix expression: ABC*+DE/- b) Evaluate the value of the postfix expression: 2, 3, *, 5, 4, *, +, 9, 6, /, 7, 2, *, -, + c) What is the difference between Longest Common Subsequence (LCS) and substring? Explain with an example. d) Generate infix expression from the following binary tree: <pre>graph TD Root((*)) --- A((A)) Root --- Plus((+)) Plus --- B((B)) Plus --- Slash(/) Slash --- C((C)) Slash --- D((D))</pre>	[1+ 4+ 2+ 3]															



PMSCS PROGRAM

Department of Computer Science and Engineering
Jahangirnagar University

Date:
05/09/2025

Course Code: PMSCS-623P

Title: Data Structures and Algorithms

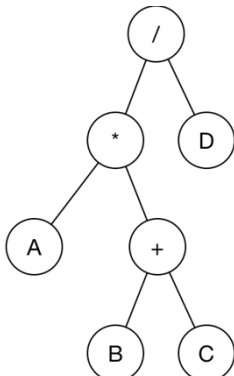
Time: 3 Hours

TERM: SUMMER 2025

Full Marks: 60

Answer any SIX from the following Questions

1.	a) Let's say you have a string "DATA". We like to print the string as it is. Which data structure will you use (stack / queue? for this task? Finally, perform the task using your selected data structure. b) A pharmacy receives medicine batches throughout the day. To prevent selling expired medicine, which data structure should you suggest along with the required operations for easy stock management? Explain briefly about your choice. c) Consider the following input: 1, 2, 3, 4, 5, 5, 4, 3, 2, 1. Should you apply normal Quicksort or Randomized Quicksort algorithm to sort the given input? Justify your choice.	[3+ 3+ 4]
2.	a) The MergeSort algorithm we have learned is called 2-way MergeSort because it divides the given array into two almost equal sub-arrays. In 4-way MergeSort, the given array is divided into 4 equal (almost) sub-arrays and they are recursively sorted and then combined/merged to get the sorted array. Represent the time complexity of 4-way MergeSort algorithm in Big-Oh notation. (Hint: The combine step merges a total of n elements, taking $O(n)$ time.) b) What is the number of comparisons required by Insertion Sort for a reverse-sorted array of size $(n-3)$? c) After the third pass of Bubble Sort, which element is guaranteed to be in its final position? d) A library management system needs to sort a list of books by their ID numbers whenever new books are added. Suppose the library has book IDs: [101, 102, 104, 103, 105]. Your task is to choose the most efficient sorting algorithm between Insertion Sort and Bubble Sort and justify the choice based on number of operations required to perform the task.	[4+ 2+ 1+ 3]
3.	a) Consider the array having elements as: [7, 7, 7, 7, 7, 7, 7]. Do you think binary search perform better than linear search algorithm in this case? Justify your answer. b) A giant pizza has 512 slices, and you want to share it equally among 4 friends. You can only cut the pizza in half at a time. Using a divide and conquer strategy, how many cuts do you need to make? c) Consider the following array representation of a max-heap as [90, 15, 10, 12, 7, 2, 5]. Is this a valid max-heap? Justify your answer. d) What is procedure to restore the property of a max-heap?	[3 + 2 + 3 + 2]
4.	a) Modern web browsers allow users to navigate between web pages using the Back and Forward buttons. The challenge is to always return the user to the most recently visited page first when the Back button is clicked. Which data structure should the browser use to solve this problem and why? b) To solve same problem, the performance of the two algorithms are: Algorithm A: $f(n) = 100n$ and Algorithm B: $g(n) = n^2$. Which is better for very large n? Which is worse asymptotically? Explain briefly.	[2+ 4+ 2+ 2]

	c) Which of the following is slower? i. $O(\log n)$ ii. $O(\log n^2)$ iii. $O(2 \log n)$ d) Given an array as $[-5, 7, 14, -27, -39, 106, -65]$. Do you think that accessing the 5th element of this array will be $O(1)$?																
5.	a) How many edges will a tree have which contains total $(4n^2 + 2\log n + 53)$ nodes? b) Consider an undirected connected graph where every edge has same weight value of 2 . What information can we get if we apply BFS (Breadth-First Search) algorithm on this graph? c) BFS will not always visit every node of a tree. (True / False) – Justify your answer. d) Can two different MSTs have the same total weight? – Justify your answer. e) What happens if one extra edge is added to an Minimum Spanning Tree?	[2+ 2+ 2+ 2+ 2]															
6.	a) If all symbols have equal frequency, is the Huffman code unique? b) Why must Huffman codes satisfy the prefix property? c) Construct the Huffman code for the following given symbols and frequencies: <table border="1"><tr><td>Symbol</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td></tr><tr><td>Frequency</td><td>45</td><td>13</td><td>12</td><td>30</td><td>12</td></tr></table>	Symbol	A	B	C	D	E	Frequency	45	13	12	30	12	[2+ 2+ 6]			
Symbol	A	B	C	D	E												
Frequency	45	13	12	30	12												
7.	a) For activity/interval scheduling problem in greedy approach, why <i>is the earliest finish time approach</i> chosen as the best strategy? (b) Consider the following coin system having coins as $\{1, 5, 6, 9\}$ and the amount that we want to exchange is 10. Do you think greedy approach will fail here? Justify your answer. c) Compute the optimal price for the 0-1 knapsack problem for knapsack capacity, W=7. Which items will be in the optimal solution? What will be their total price? Show how you calculate the optimal solution using a matrix containing both price and arrows for each cell in the matrix. <table border="1"><tr><td>Item</td><td>Weight</td><td>Value</td></tr><tr><td>1</td><td>3</td><td>4</td></tr><tr><td>2</td><td>1</td><td>1</td></tr><tr><td>3</td><td>2</td><td>3</td></tr><tr><td>4</td><td>4</td><td>5</td></tr></table>	Item	Weight	Value	1	3	4	2	1	1	3	2	3	4	4	5	[2+ 2+ 6]
Item	Weight	Value															
1	3	4															
2	1	1															
3	2	3															
4	4	5															
8.	a) How many stack pop operations are required to evaluate the postfix expression: $ABC^*+DE/-$ b) Evaluate the value of the prefix expression: $+, +, *, 2, 3, *, 5, 4, -, /, 9, 6, *, 7, 2$ c) What is the difference between Longest Common Subsequence (LCS) and substring? Explain with an example. d) Generate infix expression from the following binary tree: 	[1+ 4+ 2+ 3]															